

Lab 1

DATA 422: Advanced Data Science Methods and Ethics

Ungraded Tuesday lab (~80 min). **Paper quiz** Friday from the [Quiz 1 study bank](#) (this lab + Ch. 1).

Open in Colab

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Notebook file: [lab-01.ipynb](#).

What you will do

Review the **four steps** of supervised learning (§1.3) for logistic regression, with the math in the notebook. Fit one 2-D example the DATA 322 way (`sklearn.linear_model.LinearRegression` / `LogisticRegression` and `statsmodels Logit`), then rebuild **the same example** in PyTorch using only `nn.Sequential` + `sigmoid` + `BCELoss`.

Project (separate)

[Project 1 week 1](#) uses the wildfire table. That stone uses logits + `BCEWithLogitsLoss` (same logistic model, different wrapper).